

2. Project Objective

Objective

The primary objective of this project is to perform comprehensive API testing of the Fake Store REST API using Postman and Newman CLI. The project aims to verify that all API endpoints function correctly, return accurate data, and meet the expected business requirements of an e-commerce application.

API testing plays a crucial role in ensuring smooth communication between client applications and backend servers. Through this project, various API endpoints related to products, categories, carts, users, and authentication are tested to validate their functionality, reliability, and performance.

The project focuses on validating different HTTP methods such as GET, POST, PUT, PATCH, and DELETE by sending requests to the Fake Store API and analyzing the responses received from the server.

Specific Objectives

1. Validate API Functionality

To verify that all API endpoints perform their intended operations correctly and return the expected results based on the request sent by the client.

Examples:

- Retrieve all products successfully.
- Retrieve a specific product using Product ID.
- Add a new product.
- Update existing product details.
- Delete a product successfully.

2. Verify HTTP Status Codes

To ensure that the API returns the correct HTTP status codes for different requests.

Examples:

- 200 OK
- 201 Created
- 400 Bad Request
- 404 Not Found
- 500 Internal Server Error

Status code validation helps confirm whether the API request was processed successfully or failed due to an issue.

3. Validate Response Data

To verify that the response body contains accurate and expected data.

Validation includes:

- Product Name
- Product Price
- Product Category
- User Information
- Cart Details
- Response Structure

This ensures data integrity between the request and response.

4. Validate JSON Response Structure

To verify that the API response follows the expected JSON format and contains all required fields.

Examples:

- id
- title
- price
- category
- description
- image

This helps ensure consistency and correctness of API responses.

5. Validate Response Time

To measure API response time and ensure that responses are received within acceptable performance limits.

Performance validation helps identify slow API responses that may impact user experience.

Example:

- Response time should be less than 2000 milliseconds.

6. Perform CRUD Operation Testing

To validate Create, Read, Update, and Delete operations available in the Fake Store API.

CRUD Operations Covered:

Create:

- Add New Product
- Add Cart

Read:

- Get All Products
- Get Single Product
- Get All Users

Update:

- Update Product (PUT)
- Update Product (PATCH)
- Update Cart

Delete:

- Delete Product
- Delete Cart

7. Implement Automated API Validation

To write JavaScript-based test scripts in Postman for automatically validating responses during execution.

Automated validations include:

- Status Code Verification
- Response Time Verification
- Response Body Verification
- Data Type Validation
- Environment Variable Validation

This reduces manual effort and improves testing efficiency.

8. Manage Environment Variables

To use environment variables for storing reusable data such as:

- Base URL
- Product ID
- User ID
- Cart ID
- Authentication Token

Environment variables improve maintainability and make test execution easier across different environments.

9. Execute Collections Using Collection Runner

To execute multiple API requests together using Postman Collection Runner and verify complete workflow execution.

Benefits:

- Faster execution
- Automated validation
- Easy result analysis

10. Automate Collection Execution Using Newman CLI

To run Postman collections from the command line using Newman and support automated API testing.

Benefits:

- Command-line execution
- Continuous Integration support
- Automated report generation
- Batch execution of test cases

11. Generate Test Execution Reports

To generate detailed execution reports using Newman that provide information about:

- Total Requests Executed
- Passed Test Cases
- Failed Test Cases
- Response Times
- Error Details
- Overall Test Summary

These reports help stakeholders understand API quality and execution results.

12. Gain Practical Experience in API Testing

To develop hands-on experience in real-world API testing activities including:

- API Request Creation
- API Validation
- Test Script Development
- Collection Management
- Environment Configuration
- Automated Execution
- Report Generation
- Defect Identification

This project simulates practical industry-level API testing and helps build skills required for QA and Automation Testing roles.

Summary

The overall objective of this project is to validate the functionality, reliability, accuracy, and performance of the Fake Store REST API by using Postman and Newman CLI. The project ensures that all APIs work as expected, return correct responses, handle data properly, and support automated execution with detailed reporting, thereby demonstrating complete end-to-end API testing practices used in real-world software testing projects.